

Brooke Costin

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SUMMARY

Biomedical engineer (BS, University of Rhode Island) specializing in additive manufacturing and multi-material 3D printing. Hands-on experience in DfAM, FFF process optimization, and embedding electronics directly into printed parts, backed by a strong anatomy and physiology foundation for medical device applications.

EDUCATION

University of Rhode Island, Kingston, RI

Graduated May 2025

- Bachelor of Science in Biomedical Engineering
- Minor in Mathematics

Relevant coursework: Linear Circuits, Digital Circuits, Biomechanics, Bioelectricity, Signals & Systems, Bio-measurement, Signal Processing, Engineering Statistics, Biomaterials, Medical Imaging.

ENGINEERING PROJECTS

InSOLE | Multi-Material Additive Manufacturing & Embedded Electronics — Capstone

2024-2025

- Applied design-for-additive-manufacturing (DfAM) principles to develop a fully 3D-printed capacitive-sensing insole, embedding conductive traces directly into a flexible print for real-time gait and performance monitoring.
- Optimized FFF process parameters for flexible and conductive filaments, resolving multi-material compatibility and print-quality challenges to achieve reliable, repeatable prints.
- Iterated sensor geometry and material selection through hands-on testing, integrating Multi3D Electrifi conductive traces in a TPU substrate to produce a functional single-build multi-material sensor.

Custom ECG Design | Biomeasurements Laboratory

2023-2024

- Built an ECG amplifier and implemented QRS detection and a heart-rate meter using timer- and interrupt-driven embedded signal processing, programmed in C on a microcontroller.

Digital Design FFF Project | Foundation of Engineering II

2022-2023

- Designed an eight-component articulated LEGO figure in SolidWorks and manufactured the working assembly using FFF 3D printing.

TECHNICAL SKILLS

CAD & Design: SolidWorks, Fusion360, OnShape, Shapr3D, KICAD

Programming & Analysis: Python, MATLAB, C++, MultiSim

Hardware: Arduino | ESP32 | Capacitive Sensing (FDC1004) | Breadboarding & Soldering | Oscilloscope | Signal Conditioning

Other: CPR Certified, PCB Design, CITI Certified (RCR, Social Behavioral), ISO experience, IRB, Human Testing

EXPERIENCE

Teaching Assistant, Anatomy & Physiology II — Dept. of Biological Sciences, URI

2023-2024

- Instructed undergraduates in complex human body systems (cardiovascular, respiratory, endocrine), translating dense physiological concepts into accessible explanations for students at varied levels.
- Developed and refined course materials in collaboration with faculty, reinforcing the anatomy and physiology foundation that underpins medical device and biomechanics work.

Laboratory Assistant — Dept. of Biological Sciences, URI

2023-2024

- Prepared wet labs and dissection specimens, maintained laboratory equipment, and enforced safety protocols across recurring lab sessions.
- Executed detailed prep and handling procedures with the precision and repeatability required in a controlled lab environment.

Waitress & Bartender — Belle Isle Seafood, Winthrop, MA

2017-Present

- Balanced 40+ hours per week alongside a full engineering course load while managing daily inventory and supply ordering, demonstrating sustained time management in a fast-paced environment.